

4-DOF Haptic Rendering for Hysteroscopic Surgery Simulation

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ABSTRACT

Surgical simulation is one of the ways to train surgical skills. The haptic interaction between surgical instruments and soft tissues is one of the basic functions of surgical simulation. In this paper, we propose a 4-DOF haptic rendering method for hysteroscopic surgery simulation. The optimal positioning of a virtual tool to prevent penetration into soft tissue is determined by minimizing the elastic potential energy of the simulation system. The experimental results show that our method can produce smooth feedback force and real-time rigid-soft interaction deformation.

Index Terms: 4 DOF Haptic rendering, virtual surgery, soft body dynamics

1 INTRODUCTION

The interaction between surgical instruments and soft tissues is one of the basic and crucial functions of surgical simulation. During the rigid-soft interaction, users control the virtual tool through haptic devices. When the virtual tool collides with soft tissue, it must be kept the surface of soft tissue to avoid penetration. Meanwhile, the soft tissue should be deformed in response to the collision. The feedback force is calculated by the difference between the haptic tool and the virtual tool. The core of the haptic rendering is to calculate the non-penetration position of virtual tool. To ensure the smooth feedback force, the calculation of haptic rendering must be performed at a high frequency. Hysteroscopic surgery is an important surgical approach for treating gynecological diseases [3]. The surgeon passes the surgical instruments through the cervix into the uterus to observe and operate on the pathological tissue. Due to the constraints of the cervix, the movement of the surgical tool is restricted to the 4-DOF polar coordinates system. To realize haptic rendering in the scenes of hysteroscopic simulation, we propose an energy based four degree of freedom hysteroscopic surgery simulation method that can provide smooth feedback force and soft deformation effects. Our simulation system is displayed in Fig. 1.

2 RELATED WORKS

Haptic rendering is widely used in mechanical assembly, surgical simulation and other simulation environments. In these applications, haptic rendering should provide a realistic feedback using a peripheral devices. These devices have similar appearance as real operable tools so they can be used in training and drill of real surgery. Haptic rendering can bring great realism to surgical simulation and other fields. Several previous works [1, 4] have provided a good research base for haptic rendering. Our proposed method is energy-based and can handle 4-DOF interaction problem in hysteroscopic surgery.

Projective dynamics (PD) [2] is a simulation method for deformable material based on energy constraints projection. One of the most important characteristic of PD is that it uses a local-global solver to project the vertex to acceptable place for all the related constraints.

3 ENERGY-BASED 4-DOF HAPTIC RENDERING

In the scene of hysteroscopic surgery, the pose of the surgical tool is limited and must penetrate into the entry of vagina. We design

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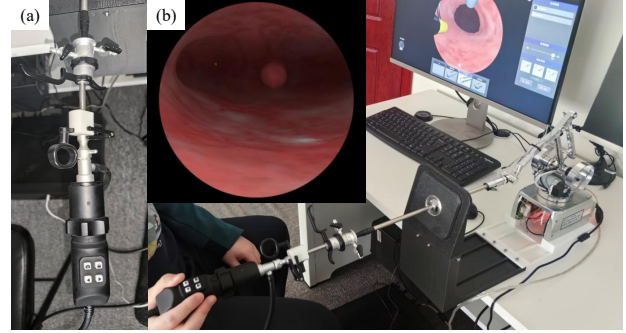


Figure 1: (a) Operation handle for our 4 DOF haptic device for hysteroscopic surgery. (b) Scene of using 4 DOF hysteroscopy simulation system, the rendering result is displayed at the upper left corner.

this limited pose of the surgical tool as a 4 degree of freedom pose $\mathbf{X} = [d, \omega]$, one degree of freedom d is for the dimension of dip in, and the other three dimensions ω for the attitude of virtual tool represented in axis-angle rotation [6]. The entry point, also the rotational center of surgical tool, is fixed in world coordinate. In the local coordinates of the virtual tool, the rotational center of the tool will move along the long axis of virtual tool.

In our simulation system, deformation part is implemented using projective dynamics [2]. In practice, the haptic end and the deformation end simply run in different frequencies. Synchronization between ends with different update rates is still a hard problem. Energy is rate-irrelevant and it is suitable for handling synchronization problem between haptic end and deformation end. Our energy-based method synchronize these two ends runs at different frequencies. In our simulation system, relationships are represented using an energy-based constraint defined based on Hooke's law: $E = \frac{1}{2}k\Delta d^2$, which is similar to the elastic potential. The constraint energy is defined as the energy projecting the point onto zero potential hyperplane following [2].

We represent our surgical tool using virtual coupling [5], while physical tool is operated by user directly and virtual tool is used for collision and interaction with soft body. Virtual coupling energy is defined as follows:

$$E_{tvc} = \frac{k_{vcs}}{2} \|d_g - d_h\|_F^2 + \frac{k_{vct}}{2} \theta(\omega_g, \omega_h)^2 \quad (1)$$

Here, d_g and d_h are the dip in depth of virtual tool tip and physical tool respectively. ω_g and ω_h are the 3-DOF axis-angle rotation of virtual tool and physical tool respectively. $\theta(\cdot, \cdot)$ is the rotation difference between the virtual tool and the physical tool. k_{vcs} and k_{vct} are the stiffness of shift energy and torsion energy respectively.

To make the virtual surgical tool stay at the pose with minimum energy, we build the following equation using Newton's method:

$$\frac{\partial E_{tool}}{\partial \mathbf{X}_g} + \frac{\partial^2 E_{tool}}{\partial \mathbf{X}_g^2} \Delta \mathbf{X}_g = \mathbf{0} \quad (2)$$

where E_{tool} is the total constraint energy on virtual tool including virtual coupling energy and contact energy. $\Delta \mathbf{X}_g = [\Delta d_g, \Delta \omega_g]$ is the

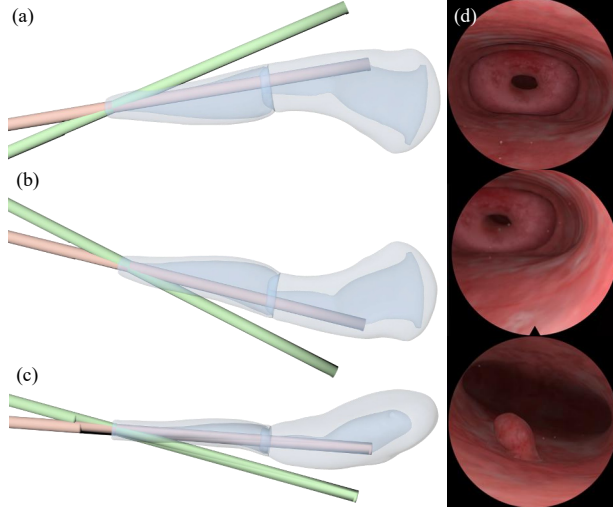


Figure 2: Deformation and rendering results of hysteroscopic surgery simulation. (a),(b) and (c) shows the surgical tool goes into the uterus and pressing left, right and down side of the uterus. Virtual tool is rendered in orange and physical tool is rendered in light green. (d) presents the images rendered inside the uterus.

update of virtual tool pose. The update of tool pose is translating the tool with Δd_g and rotate the virtual tool around the rotational center for $\Delta \omega_g$, which is represented in axis-angle fashion. The partial derivatives is described in appendix in detail.

Feedback force output to haptic device can be calculated using the partial derivative of virtual coupling energy:

$$F_{out} = \frac{\partial E_{Ivc}}{\partial d_g} \quad \mathbf{T}_{out} = \frac{\partial E_{Ivc}}{\partial \omega_g} \quad (3)$$

The calculated one dimensional force and three dimensional torque can be easily converted to the torque of twisting force on the electric motors of force feedback device.

Our method can also adapt to simpler 3 DOF haptic devices. For 3 DOF devices unable to output 4 DOF force and torque, we can simply calculate 3 DOF output force as follows:

$$\mathbf{F}_{out} = k_{vc}(\mathbf{P}_g - \mathbf{P}_h) \quad (4)$$

where $\mathbf{P}_g$ is the position of virtual tool tip and $\mathbf{P}_h$ is the position of physical tool tip.

4 EXPERIMENTAL RESULTS

Our 4 DOF haptic feedback device is displayed in Fig. 1. Physical tool is limited to the entry of vagina. Our hysteroscopic surgical tool includes a force feedback device at the tip of surgical tool, which can provide feedback force to the user through the operation lever. This force feedback device at the tip of operation lever can be a haptic device output 4 DOF feedback force or a 3 DOF haptic device. The operation lever is equipped with a set of operation keys similar to real hysteroscopic surgical device implementing functions such as changing the focal length and direction of the hysteroscope. The computer in our experiment is equipped with an Intel i7-13700 processor and a NVIDIA RTX 3090 graphics card. The scene is rendered using Unity. Rendering results of different parts in hysteroscopic surgery is displayed in Fig. 2.

We use our modified 4 DOF device to output our 4 DOF force and torque. As described in Eq. (4), we can also convert our calculated force and torque into a 3 DOF force to adapt more haptic devices such as Phantom. The calculated force is visualized in

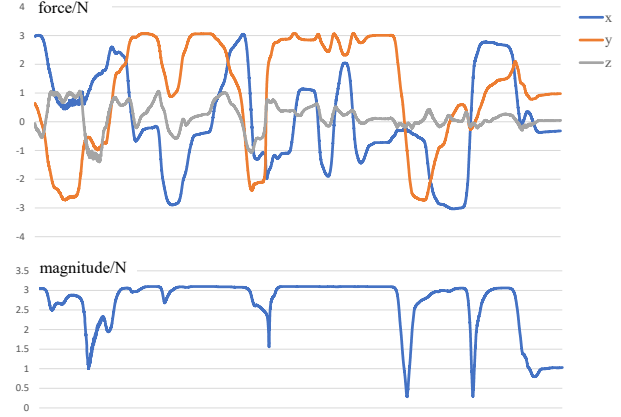


Figure 3: Visualization of output force, recording the feedback force pressing different parts in uterus. The upper part visualizes the rendered force and the lower part displays the magnitude of rendered force.

Fig. 3. The pose of the surgical tool has no sudden jump in narrow spaces and will not penetrate the uterus. The feedback force get smoother benefiting from the smooth tool pose, which will enhance the operator's sense of presence.

5 CONCLUSION

This article presents an energy-based 4 DOF haptic rendering method for virtual hysteroscopic surgery, which can also be applied on 3 DOF haptic rendering devices. Our approach meets the requirement of virtual tool pose smoothness in virtual hysteroscopic surgery, providing real-time haptic feedback force. Future enhancements will focus on exploring its application in other virtual surgical scenes such as laparoscopic liver resection surgery.

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